Md Kamruzzaman

AI & Backend Engineer | MLOps | FastAPI, Node.js | PostgreSQL, MongoDB

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GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

B.Tech Computer Science Graduate (Machine Learning, 2025) with hands-on experience building and deploying secure REST APIs and ML pipelines. Primary expertise in Python, FastAPI, and PostgreSQL, with proven adaptability to Node.js and MongoDB. Skilled in JWT authentication, database indexing, Docker, and CI/CD. Seeking Backend or MLOps roles to build reliable, high-performance systems.

TECHNICAL SKILLS

Backend & APIs: Python, FastAPI, REST APIs, SQLAlchemy, Pydantic, Node.js, Express.js
Databases: PostgreSQL, SQL, MongoDB, Mongoose, ChromaDB
Auth & Security: JWT Authentication, bcrypt Hashing, Role-Based Access Control (RBAC)
MLOps & AI: MLflow, Model Serving, Scikit-learn, LangChain, RAG Pipelines
DevOps & Tools: Docker, Kubernetes, GitHub Actions (CI/CD), Git, Render, Vercel

PRACTICAL EXPERIENCE

Backend & AI Developer (Independent Projects)

NoteVault — Secure Note Management System (Node.js & MongoDB) October 2026

Backend Code | Live Demo

- Fulfilled a backend technical assignment by successfully adapting core Python/FastAPI architectural concepts to a Node.js, Express, and MongoDB stack.
- Implemented secure API endpoints with JWT authentication, bcrypt password hashing, and Role-Based Access Control (RBAC) for User and Admin roles.
- Designed compound MongoDB indexes (`{owner: 1, createdAt: -1}`) to optimize pagination and prevent in-memory sorts, mirroring PostgreSQL optimization strategies.
- Built complex MongoDB aggregation pipelines using `$unwind`, `$group`, and `$facet` for single-call paginated queries, and `$lookup` to join User and Post collections.

Expense Tracker API — Backend Engineering & CI/CD March 2026 – October 2026

Code | Live API

- Engineered a secure REST API with **8 endpoints** using FastAPI, enforcing user-scoped data isolation.
- Designed a normalized PostgreSQL schema with SQLAlchemy and Alembic, utilizing Pydantic for strict data validation.
- Achieved **91% test coverage** with **11 automated pytest tests**, executing via GitHub Actions CI/CD.

ML Expense Platform — MLOps & Model Serving September 2026 – October 2026

Code | Live API

- Architected an end-to-end ML serving pipeline using FastAPI, scikit-learn, and MLflow to classify free-text data.
- Achieved ∼**5ms inference latency** by loading models into RAM at API startup, handling 500+ requests/second.
- Containerized the application with Docker and instrumented Prometheus metrics and Grafana dashboards.

Docu-RAG API — Generative AI & LLM Integration October 2026 – Present

Code

- Developed a Retrieval-Augmented Generation (RAG) API using FastAPI, LangChain, and ChromaDB to enable context-aware question answering over PDF documents.
- Engineered a document ingestion pipeline using `RecursiveCharacterTextSplitter` to generate and store vector embeddings.
- Integrated the Google Gemini API for semantic search, achieving a **2–4 second end-to-end latency**.

EDUCATION

Lovely Professional University, Punjab, India 2025
B.Tech in Computer Science and Engineering, Machine Learning Specialization

CERTIFICATIONS

Natural Language Processing Specialization (DeepLearning.AI) | Meta Back-End Developer (Coursera) | Python Data Structures (University of Michigan)